A COMPARATIVE ANALYSIS OF GENDER DISPARITY IN PROGRAMMING SKILLS USING EDUCATIONAL DATA MINING

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By

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Chapter 1
Literature Review

1.1 Introduction

This chapter reviews the most relevant research in the field of Educational Data Mining (EDM). Common applications include student classification, dropout prediction, detection of problematic behaviors, prediction of University Students' Academic Performance and achievement forecasting. The chapter highlights the most effective approaches in EDM that significantly impact students, and it also examines the datasets and methodologies employed in these studies.

1.2 Related works

Related work in Educational Data Mining (EDM) explores how data-driven techniques are used to understand and improve student learning and academic outcomes. Many studies focus on tasks such as student performance prediction, dropout detection, behavior analysis, engagement monitoring, and personalized learning support. Researchers commonly apply machine learning, data mining, and statistical methods to large educational datasets collected from learning management systems, assessments, and surveys. The majority of the relevant EDM research work is included in this part. EDM has become one of the most active study disciplines in recent years.

1.2.1 Forecasting the academic performance of students

The process of evaluating student performance is one of the topics that EDM researchers are most interested in studying. Murnawan et al. [1]reviewed 43 studies on Educational Data Mining, identifying student demographics (34%) and prior academic performance (26%) as the most critical factors for predicting student success. They concluded that classification is the dominant approach, with Naïve Bayes and Decision Trees serving as the most frequently used algorithms for analyzing these predictive factors.

Kumari et al. [2] and Meghji et al. [3] both identified Decision Trees as a highly effective and interpretable technique for predicting student performance, achieving accuracies of 95.20% and 96.45% respectively in their systematic reviews. While Triayudi et al. [4] found Naïve Bayes to be the most accurate (61.18%) in their specific framework, they also noted the value of Decision Trees for interpretability and the importance of including first-year data. In terms of identifying at-risk students and predicting CGPA, Ram et al. [5] achieved high precision using linear ($R^2 = 0.96$) and logistic regression

(94% accuracy). Additionally, Rahman et al. [6] demonstrated a strong link between co-curricular activities and academic success, predicting outcomes with 99.53% accuracy using Logistic Regression.

1.2.2 Forecasting the programming performance of students

Improving the computer programming skills of CSE students receives a lot of attention. Choi et al. [7] conducted a systematic literature review on performance prediction in programming education, analyzing recent trends in Educational Data Mining (EDM) over the last five years. The study highlights that while various machine learning techniques are employed, there is a significant shift towards using deep learning models and ensemble methods to improve prediction accuracy. Furthermore, the authors emphasize the growing importance of utilizing diverse data sources, such as student interaction logs and coding snapshots, rather than relying solely on assessment scores to capture the complexity of learning programming.

Mutka et al. [8] utilized real-time behavioral tracking via ”AssessMe” to identify at-risk students and improve retention. Islam et al. [9] achieved 96% accuracy in predicting programming skills using an explainable Stacking-SRDA model, pinpointing problem-solving frequency as a key factor. Chou and Chen [10] linked help-seeking behaviors to academic success, classifying patterns with 98.6% accuracy using Random Forest to aid early intervention.

1.2.3 Forecasting of student dropout

Predicting student dropout in educational institutions is one of the main applications of EDM. For educational institutions, dropout prediction is an essential duty since it may help identify at-risk students early and put preventative steps in place to keep them in institutions.

Reina Marín et al. [11] identified psychological support as the leading predictor of dropout risk—surpassing GPA—using Random Forest and XGBoost algorithms. Melo et al. [12] evaluated XAI frameworks for dropout prediction, establishing SHAP as the most effective method ($IE_{XAI} = 0.78$) for highlighting socioeconomic factors. Additionally, Agrusti et al. [13] reviewed 73 studies, finding that Decision Trees (67%) are the predominant technique used to predict university dropout, followed by Bayesian models and Neural Networks.

1.2.4 Gender differences in AI-assisted programming learning

Programming is a core component of every computer science curriculum, making the development of strong programming skills and high academic performance essential. Student performance in programming courses is evaluated through various methods, including projects, continuous assessments, and final semester examinations. Based on this, the author investigated that while both genders value programming for future careers, male students are significantly more confident and self-reliant, whereas female students perceive greater difficulty and rely more on external help [14]. To address this gap, the study recommends using visualization tools and peer instruction, noting that students surprisingly prefer traditional lectures over project-based learning.

Maurat et al. [15] found a positive correlation ($r = 0.296$) between AI tool usage and perceived effectiveness, noting that while male students used tools more frequently, females reported higher effectiveness ratings. Singha et al. [16] achieved 84.61% accuracy using the Stacking-DSAR model, revealing that male students prioritize competitive online problem solving, whereas female students rely on structured academic routines.

1.3 Problem Statements

While Educational Data Mining successfully predicts student performance using academic metrics like CGPA, current models often overlook critical non-academic factors such as psychological support and help-seeking behaviors. Moreover, existing frameworks frequently fail to account for gender disparities in programming confidence and the evolving impact of AI-assisted learning tools on student engagement. Finally, a significant gap remains in balancing high prediction accuracy with the interpretability needed for educators to intervene effectively. The following table 1.1 highlights these research gaps.

Table 1.1: Evaluation of existing EDM models based on Accuracy, DSS integration, and Explainability (XAI)

References	No. of sample used	Model	Cross-Validation	Accuracy	DSS	XAI	Observations
[4]	300	03 (DT, NN, NB)	Yes	61.18%	Yes	No	Small dataset; proposed a framework for academic guidance but achieved relatively low accuracy.
[6]	850	04 (LR, Voted Perceptron, MLP, RF)	Yes	99.53%	No	No	Investigates co-curricular activities; unusually high accuracy reported for survey data; no XAI or DSS.
[9]	1720	07 (Stacking-SRDA, RF, DT, SVM, ANN, KNN, NBC)	Yes	92.4%	Yes	Yes	Large dataset; novel Stacking-SRDA model outperformed others; integrates XAI (LIME, SHAPASH) and skill gap identification.
[16]	484	06 (Stacking-DSAR, RF, DT, SVM, ANN, KNN)	Yes	84.61%	No	Yes	Focuses on gender disparity in programming; uses XAI (SHAP, ELI5, LIME) to analyze learning behaviors
[17]	830	05 (RF, DT, NB, NN, KNN)	No	98.70%	No	No	High accuracy achieved with RF and DT; limited to a specific department and 5-6 features.
[18]	Not Specified	02 (RNN-LSTM, CFS)	Not Mentioned	97.10%	No	No	Proposed deep learning (LSTM) for time-series; used CFS for feature selection; high accuracy but sample size unclear.
[19]	177	04 (kNN, DT, LR, NN)	Yes (10-fold)	95.5%	No	No	kNN achieved highest accuracy; identified assignments/attendance as key predictors; small dataset.
[20]	651	03 (K-Means, NB, DT)	No	75.42%	No	No	Used K-means ($k = 2$) for clustering prior to classification; Naïve Bayes achieved moderate accuracy.
[21]	40 (Papers)	Various (SVM, RF, DT, LSTM, RNN, CNN)	Yes (k-FCV)	99.18%	No	No	Systematic review of 40 papers (2018-2022); identifies a lack of deep learning and XAI use in programming EDM[cite: 13, 16, 58, 302].
[22]	1720	06 (RF, SVM, DT, ANN, KNN, NBC)	Yes (10-trials)	91%	Yes	No	Real dataset from Bangladesh; RF and SVM outperformed others; includes a Knowledge-based Agent for student feedback[cite: 948, 949, 1601, 1683].
[23]	7,097	01 (Random Forest)	Yes (Preprocessing)	N/A (Focus on Explainability)	No	Yes	Brazilian HEI dataset; uses XAI (SHAP) to interpret dropout factors; identifies Coefficient of Return (CR) as the most significant feature.

1.4 Proposed Solutions

Previous studies on Student Data Mining (EDM) have demonstrated promising performance but also revealed several methodological and practical limitations. To overcome these gaps, the proposed framework incorporates Bayesian Optimization for systematic hyperparameter tuning, along with a real-time Decision Support System (DSS). In these studies, students are evaluated for programming performance without any bias, ensuring that gender does not affect the assessment process. Building on this perspective, we propose an EDM-based system to identify the key factors influencing programming success and to classify the performance of male and female students separately using Explainable AI (XAI) tools. This approach enhances usability, trust, transparency, and interpretability in real-world applications. The system's reliability and generalizability were validated through 5-fold cross-validation, leading to performance improvements over existing methods.

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